

MINOT AFB, ND, USA

WMO: 727675

Lat: 48.417N

Lon: 101.350W

Elev: 1667

StdP: 13.83

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94011

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.2	-17.2	-26.6	1.3	-21.7	-21.8	1.8	-16.5	35.0	14.1	32.4	17.7	10.4	300	0.715

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.8	90.4	69.1	86.8	68.2	83.7	66.5	73.3	84.0	71.1	81.6	69.0	79.6	12.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.2	118.3	78.9	67.7	108.5	76.7	65.6	100.5	74.7	38.1	84.4	36.0	81.8	34.2	79.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.2	26.6	23.8	-28.4	97.9	5.3	4.1	-32.2	100.8	-35.3	103.2	-38.3	105.5	-42.1	108.5
			-28.2	77.9	4.8	2.8	-31.6	79.9	-34.4	81.5	-37.1	83.1	-40.5	85.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.3	8.5	12.7	25.9	39.7	54.0	63.6	68.6	67.5	57.8	43.0	27.0	13.8
	DBStd	23.95	15.21	14.54	14.27	10.72	8.17	6.67	5.52	6.22	8.27	9.25	12.13	14.34
	HDD50	5557	1286	1045	748	331	52	3	0	0	24	255	692	1121
	HDD65	9356	1751	1465	1211	758	351	100	26	45	241	681	1141	1586
	CDD50	2032	0	0	2	23	176	410	578	544	258	39	2	0
	CDD65	357	0	0	0	0	9	58	139	124	26	1	0	0
	CDH74	4782	0	0	0	20	216	763	1600	1635	516	31	1	0
	CDH80	1631	0	0	0	3	53	220	527	648	172	8	0	0
Wind	WSAvg	11.2	12.2	11.8	11.9	12.0	12.0	10.7	9.3	9.1	10.3	11.6	11.5	11.7
Precipitation	PrecAvg	17.6	0.4	0.4	0.7	1.0	2.9	3.8	2.7	2.0	1.4	1.2	0.7	0.6
	PrecMax	23.1	1.4	3.5	1.3	2.3	6.7	10.0	7.8	3.9	4.1	5.8	2.7	1.6
	PrecMin	11.7	0.0	0.0	0.1	0.3	0.6	1.5	1.1	0.6	0.3	0.0	0.0	0.1
	PrecStd	3.4	0.4	0.7	0.4	0.5	1.6	1.6	1.4	0.9	0.9	1.2	0.7	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.6	46.4	65.9	76.8	86.3	90.9	94.1	96.7	91.0	77.7	62.7	47.9
		MCWB	37.3	39.5	50.2	55.2	63.4	70.0	72.6	69.3	67.2	58.0	49.3	39.7
	2%	DB	38.5	39.5	57.1	69.0	79.5	86.0	89.1	90.9	84.5	70.3	54.1	41.1
		MCWB	34.9	36.1	46.4	51.3	59.5	67.1	71.7	68.2	63.1	55.0	44.2	35.6
	5%	DB	35.3	36.4	49.9	63.7	75.3	81.8	85.7	86.9	80.6	64.3	48.1	37.1
		MCWB	32.7	33.7	42.7	49.0	57.0	65.0	70.6	67.0	61.7	51.2	40.6	33.1
	10%	DB	31.0	33.2	43.4	58.6	71.6	78.4	82.3	82.9	75.0	58.7	42.9	33.3
		MCWB	29.3	31.4	38.9	46.3	55.5	63.5	68.1	66.0	59.6	49.0	37.7	30.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.3	40.4	51.8	56.8	66.1	74.7	77.9	73.9	70.5	60.9	50.3	40.0
		MCDB	41.9	45.3	62.9	73.6	79.3	84.6	86.4	86.3	84.6	72.9	61.9	46.8
	2%	WB	35.2	36.6	47.4	52.9	62.8	70.4	74.7	71.4	66.7	56.3	45.2	36.2
		MCDB	38.1	39.5	55.8	65.6	74.3	79.7	84.8	83.3	79.1	67.1	52.6	40.7
	5%	WB	32.7	33.8	42.7	49.9	59.6	67.6	72.3	69.6	63.9	53.1	41.2	33.5
		MCDB	35.3	36.1	49.4	61.8	71.3	77.8	82.1	81.2	74.3	61.7	46.9	37.1
	10%	WB	29.2	31.6	39.0	47.3	57.1	65.3	70.1	67.8	61.1	50.0	38.0	30.5
		MCDB	31.0	33.2	43.6	56.9	67.9	75.6	80.1	79.6	72.4	58.2	42.7	33.2
Mean Daily Temperature Range	5% DB	MDBR	17.8	17.7	18.7	22.8	24.1	22.4	23.8	26.7	25.3	21.5	18.3	16.7
		MCDBR	19.8	19.2	28.1	32.6	31.2	28.6	28.1	33.1	34.6	31.8	27.2	20.1
		MCWBR	17.0	15.6	18.0	17.7	14.9	14.0	13.5	14.0	16.1	17.9	18.0	15.6
	5% WB	MCDBR	19.7	18.2	27.0	30.2	27.0	24.3	25.4	27.6	29.2	29.3	25.2	19.9
		MCWBR	17.2	15.0	17.9	17.5	14.4	13.7	13.9	13.8	15.2	17.4	17.1	16.0
Clear-Sky Solar Irradiance	taub	0.250	0.257	0.291	0.329	0.345	0.366	0.376	0.384	0.341	0.298	0.265	0.243	
	taud	2.402	2.424	2.450	2.400	2.408	2.394	2.382	2.337	2.428	2.526	2.494	2.447	
	Ebn at Noon	267	292	296	293	291	284	279	271	274	272	260	257	
	Edh at Noon	20	25	29	34	36	36	36	36	30	23	19	17	
All-Sky Solar Radiation	RadAvg	501	852	1236	1553	1828	1966	2094	1714	1265	793	476	352	
	RadStd	44	83	100	116	126	141	83	120	92	87	70	39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page